

A Smooth Genuine-Hessian Counterexample to the Dimension-Free Logarithmic Local-Rate Conjecture

July 14, 2026

Abstract

We construct an explicit sequence of smooth, strongly convex potentials in equilibrium-whitened coordinates for which the Gaussian natural-gradient linearized gap has the square-root, rather than logarithmic, condition-number scale. In dimension $N_\theta = d + 1$, the potentials satisfy

$$\mathbb{E}\nabla V_d(Z) = 0, \quad \mathbb{E}\nabla^2 V_d(Z) = I, \quad \alpha_d I \preceq \nabla^2 V_d \preceq \beta_d I,$$

with post-whitening condition number $\kappa_d = \beta_d/\alpha_d \asymp d$, but

$$\gamma_{\text{loc}}(V_d) \asymp d^{-1/2} \asymp \kappa_d^{-1/2}.$$

Thus $\gamma_{\text{loc}}^{-1} \asymp \sqrt{\kappa_d} \gg 1 + \log \kappa_d$. The same examples satisfy $\tau_H^2 \gtrsim \sqrt{d} \asymp \sqrt{\kappa_d}$, so they also disprove the proposed dimension-free first-Hermite Hessian bound. The Hessian field is pointwise $\nabla^2 V_d$ with $V_d \in C^\infty$; no symmetry-only surrogate, noncommutative Loewner transfer, or unproved Bellman/packet input is used.

1 Statement of the counterexample

All constants below are numerical and independent of d . The notation $a_d \asymp b_d$ means $cb_d \leq a_d \leq Cb_d$ for two such constants.

Theorem 1.1 (Smooth square-root counterexample). *There are universal constants $c, C > 0$ and $d_0 \in \mathbb{N}$ such that, for every integer $d \geq d_0$, there is a C^∞ , strongly convex potential $V_d : \mathbb{R}^{d+1} \rightarrow \mathbb{R}$ with Hessian $W_d = \nabla^2 V_d$ satisfying, for $Z \sim \mathcal{N}(0, I_{d+1})$,*

$$\mathbb{E}\nabla V_d(Z) = 0, \quad \mathbb{E}W_d(Z) = I_{d+1}. \tag{1.1}$$

If α_d and β_d are the optimal scalar lower and upper curvature constants, then

$$\alpha_d = d^{-1/2}, \quad c\sqrt{d} \leq \beta_d \leq C\sqrt{d}, \quad cd \leq \kappa_d := \frac{\beta_d}{\alpha_d} \leq Cd. \tag{1.2}$$

The positive linearized generator at $(0, I)$ obeys

$$\frac{1}{\sqrt{d}} \leq \gamma_{\text{loc}}(V_d) \leq \frac{C}{\sqrt{d}}. \tag{1.3}$$

Consequently

$$\gamma_{\text{loc}}(V_d) \asymp \kappa_d^{-1/2}, \quad \gamma_{\text{loc}}(V_d)^{-1} \geq c\sqrt{\kappa_d}, \quad (1 + \log \kappa_d)\gamma_{\text{loc}}(V_d) \longrightarrow 0. \tag{1.4}$$

Moreover, the first-Hermite Hessian coupling satisfies

$$\tau_H(V_d)^2 \geq c\sqrt{d} \geq c'\sqrt{\kappa_d}. \tag{1.5}$$

The construction is given explicitly below. Its only dimension-dependent parameter δ_d is the smallest zero in $[0, 1]$ of the continuous scalar function in (2.11); Lemma 2.1 proves that this zero exists and has the stated scale. This is a canonical one-dimensional definition, analogous to specifying a Gaussian quantile.

2 The radial-softplus Hessian

Write $z = (t, y) \in \mathbb{R} \times \mathbb{R}^d$ and define

$$r(y) := \sqrt{1 + \|y\|^2}, \quad F(s) := \log(1 + e^s), \quad p(s) := F'(s), \quad q(s) := F''(s) = p(s)(1 - p(s)). \quad (2.1)$$

Thus $0 < p < 1$ and $0 < q \leq 1/4$. For $0 \leq \delta \leq 1$, set

$$s_\delta(t, y) := r(y) - \sqrt{d} - \delta t, \quad U_\delta(t, y) := F(s_\delta(t, y)), \quad A_\delta := \nabla^2 U_\delta. \quad (2.2)$$

Let $S = \|y\|^2$ and

$$v_\delta := \nabla s_\delta = \begin{pmatrix} -\delta \\ y/r \end{pmatrix}, \quad D(y) := \nabla_y^2 r = \frac{I_d}{r} - \frac{yy^\top}{r^3}. \quad (2.3)$$

Direct differentiation gives the exact Hessian

$$A_\delta = q(s_\delta)v_\delta v_\delta^\top + p(s_\delta) \begin{pmatrix} 0 & 0 \\ 0 & D(y) \end{pmatrix}. \quad (2.4)$$

In longitudinal/transverse blocks this is

$$(A_\delta)_{tt} = \delta^2 q, \quad (A_\delta)_{yt} = -\delta q \frac{y}{r}, \quad (A_\delta)_{yy} = q \frac{yy^\top}{r^2} + p D(y). \quad (2.5)$$

Both summands in (2.4) are positive semidefinite. The eigenvalues of D are r^{-1} in tangential directions and r^{-3} in the radial direction. Hence $0 \preceq D \preceq I_d$, while $\|v_\delta\|^2 \leq 2$. It follows that

$$0 \preceq A_\delta \preceq \frac{3}{2} I_{d+1} \quad (0 \leq \delta \leq 1). \quad (2.6)$$

Let $T \sim \mathcal{N}(0, 1)$ and $Y \sim \mathcal{N}(0, I_d)$ be independent. In the next definitions, $s_\delta = s_\delta(T, Y)$ and $r = r(Y)$:

$$Q_d(\delta) := \mathbb{E} q(s_\delta), \quad (2.7)$$

$$M_d(\delta) := \frac{1}{d} \mathbb{E} \left[q(s_\delta) \frac{S}{r^2} + p(s_\delta) \left(\frac{d}{r} - \frac{S}{r^3} \right) \right]. \quad (2.8)$$

Rotation invariance in y and oddness of the mixed block imply

$$\mathbb{E} A_\delta = \text{diag}(\delta^2 Q_d(\delta), M_d(\delta) I_d). \quad (2.9)$$

Lemma 2.1 (Isotropizing parameter). *Uniformly over $0 \leq \delta \leq 1$ and all sufficiently large d ,*

$$c \leq Q_d(\delta) \leq \frac{1}{4}, \quad \frac{c}{\sqrt{d}} \leq M_d(\delta) \leq \frac{C}{\sqrt{d}}. \quad (2.10)$$

Consequently the continuous equation

$$\delta^2 Q_d(\delta) = M_d(\delta) \quad (2.11)$$

has a zero $\delta_d \in (0, 1)$. If δ_d denotes its smallest zero and

$$m_d := \delta_d^2 Q_d(\delta_d) = M_d(\delta_d), \quad (2.12)$$

then

$$cd^{-1/4} \leq \delta_d \leq Cd^{-1/4}, \quad cd^{-1/2} \leq m_d \leq Cd^{-1/2}, \quad \mathbb{E}A_{\delta_d} = m_d I_{d+1}. \quad (2.13)$$

Proof. Let

$$\mathcal{E}_d := \{|S - d| \leq 2\sqrt{d}, |T| \leq 1\}.$$

Since $\text{Var}(S) = 2d$, Chebyshev's inequality and independence give $\mathbb{P}(\mathcal{E}_d) \geq c_0 > 0$. On $\mathcal{E}_d$,

$$(\sqrt{d} - 1)^2 \leq 1 + S \leq (\sqrt{d} + 1)^2,$$

so $|r - \sqrt{d}| \leq 1$ and $|s_\delta| \leq 2$ for every $0 \leq \delta \leq 1$. Therefore $q(s_\delta) \geq \min_{|s| \leq 2} q(s) > 0$ and $p(s_\delta) \geq p(-2) > 0$ on this event. This proves the lower bound on Q_d ; the upper bound is $q \leq 1/4$.

The identity

$$\frac{d}{r} - \frac{S}{r^3} = \frac{d-1}{r} + \frac{1}{r^3} \quad (2.14)$$

shows that the second term in (2.8) is at least $c\sqrt{d}$ on $\mathcal{E}_d$. This proves $M_d(\delta) \geq c/\sqrt{d}$.

For the upper bound, the first integrand in (2.8) is at most $1/4$. Also, for $d > 2$,

$$\mathbb{E} \frac{1}{r} \leq \mathbb{E} S^{-1/2} \leq (\mathbb{E} S^{-1})^{1/2} = \frac{1}{\sqrt{d-2}}, \quad (2.15)$$

where the last equality follows by integrating the χ_d^2 density. Combining (2.14) and (2.15) gives $M_d(\delta) \leq C/\sqrt{d}$.

The map $\delta \mapsto \delta^2 Q_d(\delta) - M_d(\delta)$ is continuous by dominated convergence. It is negative at $\delta = 0$ and, by (2.10), positive at $\delta = 1$ for all sufficiently large d . Thus a zero exists. At every zero, (2.10) gives $\delta_d^2 \asymp d^{-1/2}$ and $m_d \asymp d^{-1/2}$. The last assertion then follows from (2.9). $\square$

3 Whitening and the post-whitening condition number

Fix δ_d from Lemma 2.1, abbreviate $U_d := U_{\delta_d}$ and $A_d := A_{\delta_d}$, and set

$$\alpha_d := d^{-1/2}, \quad c_d := \frac{1 - \alpha_d}{m_d}. \quad (3.1)$$

Define

$$V_d(z) := \frac{\alpha_d}{2} \|z\|^2 + c_d U_d(z) + b_d^\top z, \quad b_d := -c_d \mathbb{E} \nabla U_d(Z) = c_d \delta_d \mathbb{E} p(s_{\delta_d}(Z)) e_t. \quad (3.2)$$

The vector b_d is finite because ∇U_d is bounded; explicitly, it is a multiple of the t direction. It changes no Hessian and makes $\mathbb{E} \nabla V_d = 0$. The Hessian is

$$W_d(z) = \nabla^2 V_d(z) = \alpha_d I + c_d A_d(z). \quad (3.3)$$

By (2.13) and (3.1),

$$\mathbb{E} W_d = \alpha_d I + c_d m_d I = I. \quad (3.4)$$

Thus the construction is already in equilibrium-whitened coordinates; every condition number below is κ_{white} .

From (2.6),

$$\alpha_d I \preceq W_d \preceq \left(\alpha_d + \frac{3}{2} c_d \right) I. \quad (3.5)$$

For fixed y , $A_d(t, y) \rightarrow 0$ as $t \rightarrow +\infty$, so the optimal lower curvature constant is exactly α_d . At $y = 0$ and $t = (1 - \sqrt{d})/\delta_d$, one has $s_{\delta_d} = 0$ and the transverse block of A_d is $I_d/2$. Hence the optimal upper constant β_d satisfies

$$\alpha_d + \frac{1}{2} c_d \leq \beta_d \leq \alpha_d + \frac{3}{2} c_d. \quad (3.6)$$

Since $m_d \asymp d^{-1/2}$, this proves $\beta_d \asymp \sqrt{d}$ and $\kappa_d = \beta_d/\alpha_d \asymp d$.

The function V_d is C^∞ and α_d -strongly convex. In particular e^{-V_d} is integrable, and W_d is a genuine pointwise Hessian whose complete mixed-derivative hierarchy holds classically (and hence also distributionally).

4 The exact local Rayleigh identity

For completeness, we record the coordinate convention used for the local gap. If $X \in \text{Sym}(d+1)$ is the physical covariance perturbation, its packed coordinate is $X/\sqrt{2}$ and its Fisher–Rao norm is

$$\|(u, X)\|_\star^2 := \|u\|^2 + \frac{1}{2} \|X\|_F^2. \quad (4.1)$$

Lemma 4.1 (Compatible Bochner–Rayleigh identity). *For a smooth Hessian field $W = \nabla^2 V$ with $\mathbb{E}W = I$, the quadratic form of the packed generator in the question satisfies*

$$4\mathcal{Q}_W(u, X) = \mathbb{E}[(2u + XZ)^\top W(Z)(2u + XZ)] + \|X\|_F^2, \quad (4.2)$$

where $\mathcal{Q}_W(u, X) = \langle (u, X/\sqrt{2}), L_\star(u, X/\sqrt{2}) \rangle$. Consequently, if $W \succeq \alpha I$ with $0 < \alpha \leq 1$, then

$$\mathcal{Q}_W(u, X) \geq \alpha \left(\|u\|^2 + \frac{1}{2} \|X\|_F^2 \right), \quad \gamma_{\text{loc}}(W) \geq \alpha. \quad (4.3)$$

Proof. Expanding the displayed block operator $L_\star$ in the packed coordinate $(u, X/\sqrt{2})$ gives

$$\mathcal{Q}_W(u, X) = \|u\|^2 + u^\top T[X] + \frac{1}{2} \|X\|_F^2 + \frac{1}{4} \langle X, H[X] \rangle_F.$$

Gaussian integration by parts, together with $\partial_k W_{ij} = \partial_i W_{kj}$ and the corresponding fourth-order mixed-derivative symmetry, gives the compatible Bochner identities

$$\begin{aligned} \mathbb{E}[u^\top W X Z] &= u^\top T[X], \\ \mathbb{E}[(XZ)^\top W(XZ)] &= \|X\|_F^2 + \langle X, H[X] \rangle_F. \end{aligned}$$

Substitution proves (4.2). All integrations by parts are classical for V_d ; equivalently they follow first for compactly supported cutoffs and then by dominated convergence from the bounded Hessian.

If $W \succeq \alpha I$, then $\mathbb{E}Z = 0$ and $\mathbb{E}\|XZ\|^2 = \|X\|_F^2$ give

$$4\mathcal{Q}_W(u, X) \geq 4\alpha \|u\|^2 + (1 + \alpha) \|X\|_F^2 \geq 4\alpha \|(u, X)\|_\star^2,$$

where the last step uses $\alpha \leq 1$. □

5 Explicit inverse-gap certificate

Take

$$u = e_t, \quad X = \text{diag}(0, \lambda_d I_d), \quad \lambda_d := \frac{2\delta_d}{\sqrt{d}}. \quad (5.1)$$

Then

$$2u + XZ = (2, \lambda_d Y), \quad \|X\|_F^2 = d\lambda_d^2 = 4\delta_d^2, \quad \|(u, X)\|_\star^2 = 1 + 2\delta_d^2 \geq 1. \quad (5.2)$$

Put

$$g(Y) := \frac{\|Y\|^2}{r(Y)} = r(Y) - \frac{1}{r(Y)}. \quad (5.3)$$

Using (2.4), the Hessian energy is exactly

$$(2, \lambda_d Y)^\top A_d(2, \lambda_d Y) = q(s_{\delta_d})(-2\delta_d + \lambda_d g(Y))^2 + p(s_{\delta_d})\lambda_d^2 \frac{S}{r^3}. \quad (5.4)$$

Two elementary radial bounds suffice. First,

$$r - \sqrt{d} = \frac{S + 1 - d}{r + \sqrt{d}} \implies \mathbb{E}(r - \sqrt{d})^2 \leq \frac{\mathbb{E}(S + 1 - d)^2}{d} le 3. \quad (5.5)$$

Since $g = r - r^{-1}$ and $r \geq 1$,

$$\mathbb{E}(g - \sqrt{d})^2 \leq 8. \quad (5.6)$$

Second, (2.15) gives

$$\mathbb{E} \frac{S}{r^3} \leq \mathbb{E} \frac{1}{r} \leq \frac{1}{\sqrt{d-2}}. \quad (5.7)$$

Because $q \leq 1/4$ and $\lambda_d = 2\delta_d/\sqrt{d}$, (5.4)–(5.7) imply

$$\mathbb{E}[(2, \lambda_d Y)^\top A_d(2, \lambda_d Y)] \leq \frac{8\delta_d^2}{d} + \frac{4\delta_d^2}{d\sqrt{d-2}} \leq \frac{C\delta_d^2}{d}. \quad (5.8)$$

At the isotropizing root, $m_d = \delta_d^2 Q_d(\delta_d)$ and $Q_d(\delta_d) \geq c$. Therefore

$$c_d \mathbb{E}[(2, \lambda_d Y)^\top A_d(2, \lambda_d Y)] \leq \frac{C}{d}. \quad (5.9)$$

The remaining two terms in (4.2) satisfy

$$\alpha_d \mathbb{E}\|(2, \lambda_d Y)\|^2 = \alpha_d(4 + 4\delta_d^2) \leq \frac{C}{\sqrt{d}}, \quad \|X\|_F^2 = 4\delta_d^2 \leq \frac{C}{\sqrt{d}}. \quad (5.10)$$

Combining (4.2), (5.9), and (5.10), then dividing by (5.2), gives

$$\gamma_{\text{loc}}(V_d) le \frac{C}{\sqrt{d}}. \quad (5.11)$$

The matching lower bound $\gamma_{\text{loc}} \geq \alpha_d = d^{-1/2}$ follows from (4.3). Together with $\kappa_d \asymp d$, this proves (1.3)–(1.4).

Geometrically, the witness is not accidental. The rank-one vector in (2.4) is $(-\delta_d, Y/r)$, and its inner product with $(2, \lambda_d Y)$ is $2\delta_d(g/\sqrt{d} - 1)$. Gaussian radial concentration makes this $O_{L^2}(\delta_d/\sqrt{d})$, while the radial eigenvalue of D is r^{-3} . The affine perturbation therefore follows an approximate null direction of the *genuine* Hessian.

6 First-Hermite certificate

Let

$$B := \text{diag}(0, I_d/\sqrt{d}), \quad \|B\|_F = 1. \quad (6.1)$$

For the operator T in the question, smooth Hessian compatibility and two Gaussian integrations by parts give

$$\begin{aligned} e_t^\top T[B] &= \mathbb{E}[t \text{Tr}(BW_d)] = \mathbb{E}[(BZ)^\top W_d e_t] \\ &= -c_d \frac{\delta_d}{\sqrt{d}} \mathbb{E} \left[q(s_{\delta_d}) \frac{S}{r} \right]. \end{aligned} \quad (6.2)$$

On the event $\mathcal{E}_d$ used in Lemma 2.1, $q(s_{\delta_d}) \geq c$ and $S/(\sqrt{d}r) \geq c$ for all sufficiently large d . Hence

$$\frac{1}{\sqrt{d}} \mathbb{E} \left[q(s_{\delta_d}) \frac{S}{r} \right] \geq c. \quad (6.3)$$

Since $c_d \asymp \sqrt{d}$ and $\delta_d \asymp d^{-1/4}$, (6.2)–(6.3) yield

$$\tau_H^2 \geq |e_t^\top T[B]|^2 \geq c\sqrt{d}. \quad (6.4)$$

Because $\kappa_d \asymp d$,

$$D_{\kappa_d} = \left(4 + \frac{4}{\sqrt{\pi}} \right) (1 + \log \kappa_d) = O(\log d) = o(\tau_H^2).$$

This proves (1.5) and directly refutes the intended dimension-free first-Hermite estimate.

Corollary 6.1 (The affine weighted Korn frontier also fails). *Put $\tilde{A}_d = W_d - \alpha_d I = c_d A_d$. Then $\mathbb{E}\tilde{A}_d = (1 - \alpha_d)I$ and $0 \leq \tilde{A}_d \preceq L_d I$ with $L_d \asymp \sqrt{d}$. For the witness (5.1),*

$$\mathbb{E}[(2u + XZ)^\top \tilde{A}_d (2u + XZ)] \leq \frac{C}{d}, \quad (1 - \alpha_d) \|X\|_F^2 \leq \frac{C}{\sqrt{d}}.$$

Thus multiplying their sum by $1 + \log(eL_d/(1 - \alpha_d)) = O(\log d)$ still gives $o(1)$, whereas $(1 - \alpha_d)\|u\|^2 \rightarrow 1$. Hence the proposed affine weighted Korn inequality cannot hold under the stated assumptions, even for smooth genuine Hessians.

7 Guardrail audit

1. The construction never replaces Hessian compatibility by symmetry of the first Hermite tensor. Equation (2.4) is the pointwise Hessian of the displayed C^∞ potential.
2. No commutation-free Loewner transfer is used. The inverse-gap certificate is the scalar Rayleigh computation (5.4)–(5.11).
3. Spectral bounds and isotropic mean are not treated as sufficient abstract data. The small energy comes from the explicit compatible radial and mixed Hessian blocks in (2.5).
4. Whitening is exact in (3.4). The constants $\alpha_d, \beta_d, \kappa_d$ are post-whitening constants; no original-to-whitened conversion is invoked.

5. The potential is C^∞ , its Hessian is bounded, and all Gaussian integrations by parts and mixed-derivative identities are classical. They also remain valid in the weak/distributional formulation.
6. The construction is genuinely high-dimensional and gives $\gamma_{\text{loc}}^{-1} \asymp \sqrt{\kappa}$, so it does not conflict with the sharp one-dimensional logarithmic examples.
7. The result concerns only the linearized local gap. It makes no claim about a nonlinear attraction radius.

Conclusion. The family (3.2) is an admissible option-(B) counterexample. The dimension-free logarithmic local-rate conjecture is false under the assumptions in the question, and the already-known square-root fallback has the correct order on this smooth family.